PRACTICE EXAM 30 / EXAM SET 05

Domain 2: Anatomy and Physiology

8 scored questions	4 answer choices per question	Answer key + explanations included

Instructions

Choose the single best answer (A, B, C, or D) for each question. Mark your responses before checking the answer key and detailed explanations at the end of this document. These practice questions are not official NHA examination items.

QUESTIONS

Q01 | Body Structures and Organ Systems

> **SCENARIO** A patient with a provider-documented cerebellar injury is beginning outpatient rehabilitation for poorly coordinated movements. The family asks why the patient can move an arm but has difficulty directing it smoothly toward a cup. The medical assistant reinforces the provider's explanation.

Which normal cerebellar function best explains this difficulty?

A. Producing the hormones that regulate skeletal muscle growth.
B. Generating the nerve signals responsible for skin pain perception.
C. Controlling the language processes required to name an object.
D. Coordinating the timing and precision of voluntary movements.

Q02 | Pathophysiology and Disease Processes

> **SCENARIO** An older adult with provider-diagnosed shingles recalls having chickenpox in childhood and asks whether the current illness must be a completely new virus. The medical assistant reinforces the provider's explanation of the relationship between the two illnesses.

Which process causes shingles in this situation?

A. A new bacterial infection develops within a hair follicle.
B. An immune reaction to a new skin product damages the skin surface.
C. Varicella-zoster virus that remained dormant in nerve tissue becomes active again.
D. A new influenza infection spreads from the respiratory tract into the skin.

Q03 | Body Structures and Organ Systems

> **SCENARIO** A patient with provider-diagnosed slow-transit constipation asks why stool can become harder when it remains in the colon longer. The medical assistant uses the provider's digestive-system handout to explain the relevant normal function.

Which normal colonic process helps explain the firmer stool?

A. Absorption of water from material passing through the colon.
B. Secretion of bile into material passing through the colon.
C. Release of digestive acid into material passing through the colon.
D. Production of insulin in response to material passing through the colon.

Q04 | Pathophysiology and Disease Processes

> **SCENARIO** A patient has provider-confirmed carpal tunnel syndrome and numbness mainly in the thumb, index finger, and middle finger. The patient asks whether the symptoms mean that the wrist bones have stopped producing sensation. The medical assistant reinforces the documented disease explanation.

Which mechanism accounts for the symptoms of this condition?

A. Inflammation destroys the articular cartilage covering the wrist joint surfaces.
B. Compression of the median nerve at the wrist disrupts sensory signaling.
C. Compression of the ulnar nerve at the elbow disrupts sensory signaling.
D. Reduced bone mineral density in the wrist prevents sensory signaling.

Q05 | Body Structures and Organ Systems

> **SCENARIO** At a routine visit, a patient asks why even a shallow skin scrape can interrupt the body's protective surface. The provider has described the scrape as limited to the outer skin layer, and the medical assistant uses a skin diagram to reinforce the explanation.

Which layer forms the outer protective surface described?

A. Dermis
B. Epidermis
C. Subcutaneous fat
D. Deep fascia

Q06 | Body Structures and Organ Systems

> **SCENARIO** A patient reviewing a provider's elbow range-of-motion instructions thinks "extension" means lifting the whole arm out to the side. The medical assistant clarifies the movement term before the provider reviews the exercise plan.

Which movement is elbow extension?

A. Bending the elbow to bring the hand toward the shoulder.
B. Turning the forearm so the palm faces upward.
C. Straightening the elbow to increase the angle between the upper arm and forearm.
D. Moving the upper arm away from the side of the trunk.

Q07 | Pathophysiology and Disease Processes

> **SCENARIO** A patient with a provider-diagnosed peptic ulcer has an order for an H. pylori stool antigen test. The patient asks whether the test measures how much acid the stomach makes. The medical assistant reinforces the provider's explanation of the test's purpose.

What is the primary purpose of this ordered test?

A. To measure the amount of acid secreted by the stomach lining.
B. To measure bleeding from the ulcer by detecting hidden blood in stool.
C. To determine whether reduced pancreatic enzymes are causing fat malabsorption.
D. To look for evidence of H. pylori infection, a possible cause of the ulcer.

Q08 | Pathophysiology and Disease Processes

Why can a low hemoglobin concentration reduce the blood's oxygen-carrying capacity even when gas exchange in the lungs is normal?

A. There is less hemoglobin available to bind and transport oxygen.
B. There is less albumin available to transport oxygen between blood cells.
C. There are fewer platelets available to carry oxygen to body tissues.
D. There are fewer white blood cells available to release oxygen from plasma.

ANSWER KEY & EXPLANATIONS

Practice Exam 30 | Domain 2: Anatomy and Physiology | 8 scored questions

01 D	02 C	03 A	04 B	05 B	06 C
07 D	08 A				

DETAILED EXPLANATIONS

Q01 | Correct answer: D

Coordinating the timing and precision of voluntary movements.

The cerebellum helps coordinate movement, posture, and balance. A person may retain the ability to move a limb yet have difficulty making the movement smooth and accurate when cerebellar function is impaired. This connects central nervous system processing with controlled skeletal muscle activity rather than with hormone production or language.

Q02 | Correct answer: C

Varicella-zoster virus that remained dormant in nerve tissue becomes active again.

Shingles is caused by reactivation of varicella-zoster virus, the virus responsible for chickenpox. After the initial infection, the virus can remain dormant in nerve tissue and reactivate later. Bacterial follicle infection and contact reactions have different causes and do not explain this established diagnosis.

Q03 | Correct answer: A

Absorption of water from material passing through the colon.

The colon absorbs water from intestinal contents as stool forms. More time in the colon can allow additional water removal, contributing to harder stool. Bile is made by the liver, acid is secreted in the stomach, and insulin is produced by pancreatic cells; these are not the colonic process described.

Q04 | Correct answer: B

Compression of the median nerve at the wrist disrupts sensory signaling.

Carpal tunnel syndrome results from compression of the median nerve at the wrist. This nerve provides sensation to the thumb, index finger, middle finger, and part of the ring finger, so compression can cause numbness in that distribution. Ulnar nerve compression involves a different nerve and commonly affects a different sensory distribution.

Q05 | Correct answer: B

Epidermis

The epidermis is the outer layer of skin and forms a protective interface with the environment. A superficial scrape can disrupt this layer even when deeper tissue is not injured. The dermis lies beneath the epidermis, while subcutaneous fat and deep fascia are located deeper still.

Q06 | Correct answer: C

Straightening the elbow to increase the angle between the upper arm and forearm.

Elbow extension straightens the joint and increases the angle between the upper arm and forearm. Flexion bends the elbow, while supination turns the forearm so the palm faces upward. Moving the upper arm away from the trunk describes shoulder abduction rather than elbow extension.

Q07 | Correct answer: D

To look for evidence of H. pylori infection, a possible cause of the ulcer.

An H. pylori stool antigen test looks for bacterial antigens that indicate this infection. H. pylori is a common cause of peptic ulcers, so identifying infection addresses a possible underlying cause. Occult blood testing checks for bleeding, while gastric acid measurement and pancreatic testing answer different questions.

Q08 | Correct answer: A

There is less hemoglobin available to bind and transport oxygen.

Hemoglobin in red blood cells binds and carries most oxygen transported in blood. A lower hemoglobin concentration reduces the available oxygen-carrying capacity even if the lungs exchange gases normally. Albumin, platelets, and white blood cells have other primary functions and do not replace hemoglobin as the main oxygen carrier.

End of Practice Exam